

A Novel Classifier Ensemble Method Based on Subspace Enhancement for High-Dimensional Data Classification

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Abstract—High-dimensional small-size data seriously affects the performance of classifiers. By combining classifiers, ensemble learning obtains higher accuracy and more robust predictions. However, these classifier ensemble methods suffer from several limitations: 1) ensemble with sample space suffers from noise and redundant features; 2) constructing sample subspace on small-size data leads to an insufficient description of sample space; 3) ensemble with random feature subspace leads to information loss, which will degrade the performance of classifiers; 4) most ensemble methods implement directly on the original feature space, which is defective in high-dimensional data with redundant and noisy features. To overcome the above limitations, a new classifier ensemble method based on subspace enhancement (CESE) is proposed for high-dimensional data classification. First, a superior subspace enhancement scheme (SSE) is designed to effectively implement feature selection and transformation for high-dimensional data, followed by generating multiple superior feature subspaces with diversity and discrimination, which enhances the representative ability of features. Second, we develop a mixed space enhancement process (MSE) based on multiscale rotation reconstruction and various subspace enhanced features of SSE. By using MSE, an effective feature fusion is constructed to obtain more diverse features. Furthermore, to improve the capability of our method, we design various feature combination strategies for enhanced features from both SSE and MSE. Comparative results on 33 high-dimensional data sets indicate that our approach CESE outperforms different mainstream integrated systems.

Index Terms—Ensemble learning, feature transformation, subspace enhancement, high-dimensional data, classification



1 INTRODUCTION

THE classification of high-dimensional data widely exists in real-world applications of pattern recognition and machine learning, such as image recognition [1], [2], text classification [3], [4], bioinformatics mining [5], and microarray gene expression data analysis [6], [7], [8]. In high-dimensional data classification, a large amount of noise and redundant features make performance of methods severely negatively affected [9], [10]. Therefore, it is a valuable challenge to effectively classify data with high feature dimension and small sample size [11], [12]. To deal with the problems caused by high-dimensional small-size data, feature selection, and transformation methods have received intensive attention [13], [14], [15], whose goal is to obtain more compact and effective feature representations. Using these effectively processed features, the complexity of computation and storage can be reduced, which contributes to improving performance of the classifier.

By adopting the search mechanism, feature selection can find a smaller subspace to represent the entire feature space [16]. In general, feature selection can be divided into filter, wrapper and embedding. The filter method performs feature selection before training the classifier, which obtains the subspace based on the evaluation score of each feature and the number of reserved features. Among them,

different types of evaluation indicators have been proposed, such as mutual information [17], information gain [18], correlation [19], and variance [20]. Unlike the filter method, the wrapper method takes the final performance of the classifier as the evaluation criteria of feature subspace [21]. This means that the wrapper method selects a subset of features that are more beneficial to performance. Although the wrapper is better than the filter one, it has to train classifiers repeatedly, resulting in large computational cost [22], [23]. The embedding method is the integration of feature selection and classifier training, both of which are completed in the optimization process [24]. Mainstream embedding methods include linear-based model [25], [26], [27] and tree-based model [28], [29], [30], [31]. Selecting crucial feature subspaces can effectively mitigate impacts of noise and redundancy in high-dimensional data, which encounters the risk of information loss.

Feature transformation focuses on mining relationships between features and mapping them into a low-dimensional space to achieve dimension reduction. It can be categorized as unsupervised methods and supervised methods. Principal component analysis (PCA) is a well-known unsupervised method [32] which has been successfully applied in high-dimensional data classification task [33], [34]. The direction with the largest variance is considered as the principal component in PCA and retains the maximum feature information [35]. Considering the label information, linear discrimination analysis (LDA) makes the distance between different classes as large as possible and the same class as small as possible in the projection space [36], [37].

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However, some researches demonstrate that in high-dimensional data, performance of LDA is as bad as random guessing [38], [39], due to the accumulation of noise. To automatically extract abstract features, different neural networks have been constructed, such as multilayer perceptron (MLP), extreme learning machine (ELM) [40], and broad learning system (BLS) [41]. Although neural network-based methods exhibit superior performance of feature transformation, they require a large training data to alleviate the overfitting [42], [43]. Therefore, the above feature transformation methods are not competent for classification of high-dimensional small-size data.

By constructing and combining multiple classifiers, ensemble learning can obtain higher accuracy and more robust prediction results [44], [45], [46], [47], [48]. Generally, the performance of individual classifiers and the diversity between classifiers are two pivotal elements that affect the overall performance of the integrated system [49], [50]. To maintain the diversity of ensemble system, several illustrious methods use different training sets to build base classifiers, such as bagging [51], boosting [52], random subspaces [53], random forest [54], rotation forest [55] and extended space forest [56]. However, in the setting of high-dimensional data classification, these methods have limitations. For methods based on sample space, such as bagging and AdaBoost [57], they cannot provide strong diversity for data with redundant or similar samples [58], and they are susceptible to noise and redundant features. For methods based on feature space, such as random subspaces and random forest, small-size feature subspace cannot guarantee sufficient features, which may lead to performance degradation of classifier [59]. Since the rotation process needs to save all components, noise can be transmitted by this process, which makes rotation forest susceptible to noise [60]. Although extended space forest can create more diverse features, its operators are executed in the original feature space [56]. Therefore, it is also affected by noise and redundant features. Most integrated systems only improve the diversity and performance of base classifiers and neglect implications for the integrated system [61]. However, some redundant or bad base classifiers may be generated in the classifier ensemble model, which will seriously affect the overall prediction performance.

To overcome the above limitations, we propose a new classifier ensemble method based on subspace enhancement (CESE), which aims at generating accurate and diverse base classifiers for high-dimensional data classification. For each base classifier in CESE, the original features are first randomly shuffled to obtain new sets with different orders. Its main purpose is to increase diversities between base classifiers. To perform feature selection and transformation for high-dimensional data, superior subspaces enhancement process (SSE) is designed. With the advantages of random tree, SSE obtains multiple superior feature subspaces with diversity and discrimination, which can alleviate the effect of noise or redundant features. Then LDA is adopted to extract more compact and effective features in each superior feature subspace, which can enhance the expression ability of features to improve the power of the model. We next develop a mixed space

enhancement process (MSE) based on various subspace enhanced features of SSE to obtain diverse and creative features. Specifically, multiscale rotation reconstructions are applied to obtain more powerful representation ability via feature fusion. Furthermore, different feature combination strategies are designed for enhanced features of SSE and MSE, and it is applied to construct more different and effective feature sets, which can improve the capacity of our model. Experiments conducted on real-world high-dimensional data sets demonstrate that the proposed CESE outperforms integrated systems with feature space over 29 out of 33 data sets, integrated systems with sample space over all the data sets, and integrated systems with hybrid strategies over 28 out of 33 data sets, respectively.

The contributions of this paper are summarized as follows: 1) for the invalidation of classifiers in high-dimensional features, we propose a superior subspaces enhancement (SSE) to effectively perform feature selection and feature transformation, obtaining multiple superior feature subspaces with diversity and discrimination. LDA with sample subset is used to reduce dimension in each superior feature subspace for extracting more compact and diverse subspace enhanced features. SSE has high flexibility since various feature transformation strategies can be combined easily; 2) to construct effective feature fusion, we develop a mixed space enhancement method (MSE) based on multiscale rotation reconstruction and the subspace enhanced features of SSE to obtain more diverse and powerful features; 3) We design various feature combination frameworks for the enhanced features from SSE and MSE to improve the capability of our method. We investigated the enhancement brought by different feature combination frameworks for our method; 4) We adopt the Kappa-Error diagram to compare the diversity and accuracy of CESE and its competitors; 5) to verify the superiority of CESE, it is compared with mainstream integrated systems on real-world high-dimensional data.

2 RELATED WORK

The classifier ensemble system uses multiple classifiers to improve the generalization [62], [63], [64]. Since it is hard to train an optimal classifier on high-dimensional data, ensemble learning becomes a feasible solution to alleviate the limitation [48], [12]. The diversity between base classifiers is crucial to performance of an integrated system. There are strategies to build different training sets to increase the diversity of base classifiers, which can be divided into 1) ensemble with deleting or weighting samples; 2) ensemble with deleting or rotating features; 3) ensemble with extending features, and 4) ensemble with hybrid strategies.

The first strategy explores diverse training sets by deleting or weighting samples, which mainly include bagging [51] and boosting [52]. Bagging constructs training subset for each base classifier by resampling the original training set with replacement [51]. It reduces the variance of base classifiers by introducing randomness to construct an ensemble model. Boosting improves performance of weak classifiers by weighting samples [52]. Adaboost assigns higher weights to samples that are misclassified in previ-

ous base classifiers [57]. Banfield *et al.* [65] evaluate classifier ensemble methods on 57 public data sets, which indicate that boosting and random forest are statistically superior to bagging. Galar *et al.* [66] propose Eusboost for imbalanced data classification, which combines evolutionary under-sampling with boosting algorithm. To maintain the diversity and reduce the spread of noise, Xu and Huang [47] develop a bagging-based SSEOS to optimize the sample distribution for each subset. However, these methods with deleting or weighting samples suffer from noise and redundant features in high-dimensional data classification. Besides, the integrated system with deleting samples leads to an insufficient description of sample space, resulting in poor classification accuracy.

The second strategy creates diverse training sets by deleting or rotating features. Similar to bagging, random subspace [53] applies subset extraction in the ensemble model. Different from bagging, it executes subset extraction in feature space. For random subspace [53] and random forest [54], each base classifier is trained with different feature subspaces. Rodriguez *et al.* [55] design a classifier ensemble method rotation forest, which applies PCA on feature subspace to generate rotation features. This method builds accurate and diverse classifiers by retaining all the principal components. Extremely randomized trees method increases the diversity of the integrated system by randomly selecting cut-point in each random subspace [67]. Stiglic *et al.* [68] propose an extended version of rotation forest rotated random forest (RoRF), which improves performance of classifier ensemble by incorporating feature extraction. Considering advantages of random subspaces and rotation method, Xia *et al.* [69] combine them with ELM and propose RSELM and RoELM, respectively. Bryll *et al.* [70] develop an attribute bagging (AB), which applies the wrapper to feature subset selection, and improves the accuracy and stability of the integrated system. Noise and redundant features can be propagated through the rotation process, which makes rotation-based methods vulnerable. Although feature subspace alleviates the problems caused by high-dimensional data, small-size feature subspace cannot guarantee a sufficient number of features, which may impair the decision-making ability of classifiers.

Extending original features to generate effective features is another strategy to increase diversities of an integrated system. Extended space forest is proposed by Amasyali and Ersoy [56]. Each base classifier is trained with original features and extended features which are generated by applying difference operators to two paired original features [82]. Random forest with ensemble of feature spaces (RF-ensemble) [71] is a random forest variant, which performs PCA or LDA to generate new features in each base classifier. Xia *et al.* [72] propose rotation random forest with kernel principal component analysis (RoRF-KPCA) to create new features that are merged and fed into a random forest classifier. Schlar and Rokach [73] use random projection to create features since randomness can provide diversity and reduce computational complexity for the construction of an ensemble model. However, the above approaches suffer from noise and redundant features because features are extended in the original feature space.

Some researches focus on combining multiple strategies to obtain diverse base classifiers. Zhang and Zhang [74] combine rotation forest and AdaBoost. Zhang and Zhang [75] integrate bagging and PCA to develop a classifier ensemble method. In particular, for the construction of each base classifier, PCA is performed on the out-of-bag samples, and new features are generated from the corresponding bootstrap samples. RSM-IPCS utilizes random subspaces to explore the synergy between the original space and principal component space [76]. To encourage capacity and diversity of base classifiers, Xu *et al.* [48] take into account high-dimensional features and small-size samples and develop an adaptive integrated system. Xia *et al.* [77] adopt bagging, boosting, random subspaces, and rotation strategies to extend random forest-based ensemble model to further improve the classification performance. Yu *et al.* [78] design a hybrid incremental ensemble learning method for noise data classification by considering the feature and sample simultaneously. Yu *et al.* [46] present an integrated system with adaptive feature selection and auxiliary training set to handle semi-supervised high dimensional data classification. Based on fuzzy relative transformation, Yu *et al.* [79] design an RS-based ensemble framework for the dataset with noisy features. Wei *et al.* [100] extract consistent features and complementary features of multimodal information and integrate them to improve expressive ability. To illustrate the diversity of classifier ensemble models, Kuncheva [80] apply kappa-error diagrams to effectively measure them.

However, most researches only consider the generation of base classifiers with diversity and are implemented directly on the original feature space, which is defective in high-dimensional data with redundant and noisy features. Therefore, these methods still have limitations when facing classification on high-dimensional small-size data.

3 A NOVEL CLASSIFIER ENSEMBLE METHOD BASED ON SUBSPACE ENHANCEMENT

Fig. 1 shows the overall framework of our classifier ensemble based on subspace enhancement (CESE). Let $\mathbf{X}^{train} = \{(\mathbf{x}_1, \mathbf{y}_1), (\mathbf{x}_2, \mathbf{y}_2), \dots, (\mathbf{x}_n, \mathbf{y}_n)\}$ be the training set, where $\mathbf{x}_i$ ($i \in \{1, 2, \dots, n\}$) denotes a training sample and $\mathbf{y}_i$ is its label. Each training sample $\mathbf{x}_i$ consists of L features. The testing set is denoted by $\mathbf{X}^{test}$. Let $\mathbf{D} = \{D_1, D_2, \dots, D_k\}$ denote base classifiers in the classifier ensemble model, and the number of base classifiers is k . To construct accurate and diverse base classifiers for high-dimensional data classification, two subspace enhancement processes are designed, including superior subspaces enhancement (SSE) and mixed space enhancement (MSE). Considering the role of diversity and effectiveness, we design multiple feature combination strategies for enhanced features of SSE and MSE to improve the capacity of our ensemble model.

Motivations of our proposed methods are as follows:

1) To effectively perform feature selection and transformation for high-dimensional data, a superior subspace enhancement (SSE) is designed. In SSE, it first shuffles feature sets randomly and applies the random tree for feature selection. With random tree, SSE obtains multiple superior

feature subspaces with both diversity and discrimination, which can alleviate effects of noise or redundant features. Then LDA with sample subset is used to reduce dimension in each superior feature subspace for extracting more compact and diverse subspace enhanced features. Therefore, SSE can enhance the expression ability of features to improve the power of the ensemble model.

2) To generate more diverse and powerful enhanced features, a mixed space enhancement method (MSE) is developed. MSE performs multiscale rotation reconstruction on subspace enhanced features obtained by SSE. Compared to original feature space, subspace enhanced features from SSE are compact and effective. By using MSE, univariate features are fused to generate multiple multivariate features which can encourage diversity and individual accuracy within the ensemble system simultaneously.

3) Considering the diversity and effectiveness, feature combination is adopted to construct more different and effective feature sets. To further improve the capability of our method, several feature combination strategies based on enhanced features from SSE and MSE are designed for the base classifiers of CESE. To better explore the enhancement brought by these strategies, various feature combination structures are designed.

With the above subspace enhancement processes, a robust and accurate integrated system can be constructed.

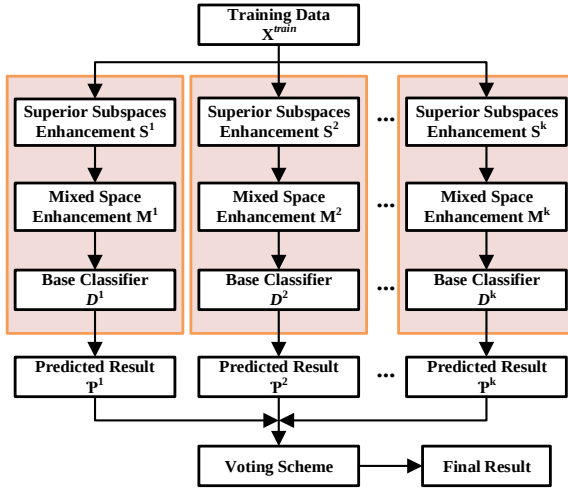


Fig. 1. The overall framework of CESE.

3.1 Superior Subspaces Enhancement

To generate accurate and diverse base classifiers, we design a new feature transformation method for high-dimensional data, referred to as superior subspaces enhancement (SSE). Studies indicate methods that perform feature transformation on high-dimensional data directly are as bad as random guessing [39]. It is necessary to select important features before feature transformation. However, it is difficult to select an optimal feature subset. To solve this problem, SSE adopts random tree [28] to obtain multiple superior feature subspaces with diversity and discrimination.

As shown in Fig. 2, for each base classifier, SSE first shuffles original features to obtain new set $\mathbf{X}^i$ ($i \in \{1, 2, \dots, k\}$) with different orders. Its goal is to increase diversities between base classifiers. $\mathbf{X}^i$ is fed into different random trees

$\mathbf{T}^i$ for training. The reason for using random tree instead of decision tree is that the former applies bagging and random subspaces, which can promote generalization.

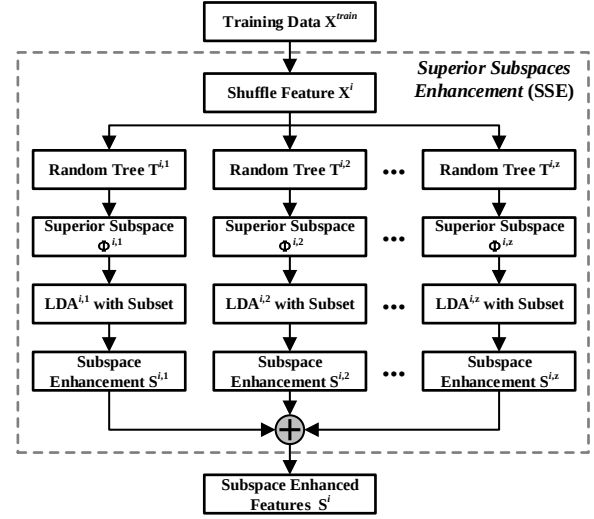


Fig. 2. The detailed structure of the SSE.

The gini index is an effective evaluation criterion for purity in random tree [81]. The importance of features is represented by $\mathcal{V}$, and the gini index is represented by $\mathbf{GI}$. The gini index of the j -th node is calculated as follows:

$$\mathbf{GI}_j = 1 - \sum_{c=1}^C p_{j,c}^2, \quad (1)$$

where C is the total number of classes and $p_{j,c}$ is the probability of class c in the j -th node. The importance of the i -th feature in the j -th node can be evaluated by the change of gini index in node partition as below:

$$\mathcal{V}_{i,j} = \frac{w_l}{w} \mathbf{GI}_j - \frac{w_l}{w} \mathbf{GI}_l - \frac{w_r}{w} \mathbf{GI}_r, \quad (2)$$

where $\mathbf{GI}_l$ and $\mathbf{GI}_r$ denote gini indices of two new nodes after the j -th node is divided. $\mathcal{W}_j$ is the number of samples in the j -th node and $\mathcal{W}$ is the number of all samples. Suppose $\mathcal{N}$ is a set of partition nodes in a random tree. The importance of the i -th feature in a random tree is defined as:

$$\mathcal{V}_i = \sum_{j \in \mathcal{N}} \mathcal{V}_{i,j}. \quad (3)$$

Each random tree generates a set of feature importance, and SSE selects the top- q features based on the importance scores to form a new superior subspace Φ . This means that the size of each superior subspace is equal to q . The size of the search subspace for each random tree $\mathcal{r}$ is a hyper-parameter. A smaller search subspace can make superior subspaces more diverse, while a larger search subspace can enlarge the similarity between a pair of superior subspaces. Although a larger search subspace can consider a wider range of features at each partition node, it can inevitably cause redundancy between superior subspaces.

To obtain more compact and effective features, a subspace enhancement process is carried out on each superior subspace. SSE concatenates features to generate subspace enhanced features $\mathbf{S}$ after subspace enhancements whose goal is to retain discriminative information in the dimension reduction. Suppose ϕ and C denote the multiple size

in SSE and the number of classes, the number of features in each superior subspace and the number of random trees in each SSE can be defined by q and z , as follows:

$$q = \varphi * (C - 1), \quad (4)$$

$$z = L / q, \quad (5)$$

where L denotes the number of features in original feature space. After subspace enhancement process, the size of features is reduced to $(C - 1)$. Similarly, φ is an important hyper-parameter in SSE. A small φ encourages superior subspaces Φ to be more diverse at the cost that effective information contained in each superior subspace may not be reduced. A large φ may preserve noise or redundant features, resulting in a decrease in accuracy. Besides, a large φ reduces diversities between superior subspaces Φ .

To increase diversity of subspace enhanced features between base classifiers, SSE uses bootstrap to produce sample subsets $\mathcal{D}$ in each subspace enhancement process. The resampling rate of bootstrap is set to 80%. $\mathcal{D}_{i,j}$ denotes the training subset for the j -th superior subspace of the i -th base classifier. To ensure that subspace enhanced features have the largest inter-class distance and the smallest intra-class distance, SSE applies LDA in each subspace enhancement process to generate subspace enhanced features. Suppose $\mathbf{S}^i$ ($i \in \{1, 2, \dots, k\}$) is the subspace enhanced features set of the i -th base classifier, and it can be defined by:

$$\mathbf{S}^i = [\mathbf{S}^{i,1}, \mathbf{S}^{i,2}, \dots, \mathbf{S}^{i,z}], \quad (6)$$

$$\mathbf{S}^{i,j} = \tilde{\mathbf{w}}_{i,j}^T \mathcal{D}_{i,j}, \quad j \in \{1, 2, \dots, z\}, \quad (7)$$

where z denotes the number of subspace enhancements. $\tilde{\mathbf{w}}_{i,j}$ is a matrix of $q \times d$, where q and d denote the sizes of each superior subspace and low-dimensional space after LDA, respectively. By using feature transformation based on LDA, SSE obtains more compact and effective features. Suppose $\mu_{i,j,c}$ denotes the mean of class c in the j -th superior subspace of the i -th base classifier. $\mu_{i,j}$ denotes the mean of data in the j -th superior subspace of the i -th base classifier. $N_{i,j,c}$ denotes the number of samples in $\mathcal{D}_{i,j}$ belonging to class c . Thus, $\mu_{i,j,c}$ and $\mu_{i,j}$ can be defined as:

$$\mu_{i,j,c} = \frac{1}{N_{i,j,c}} \sum_{\mathcal{D}_{i,j} \in c} \mathcal{D}_{i,j}, \quad (8)$$

$$\mu_{i,j} = \frac{1}{N_{i,j}} \sum \mathcal{D}_{i,j}. \quad (9)$$

The mean of class c in the j -th subspace enhancement of the i -th base classifier can be expressed as:

$$\tilde{\mu}_{i,j,c} = \frac{1}{N_{i,j,c}} \sum_{\mathcal{D}_{i,j} \in c} \tilde{\mathbf{w}}^T \mathcal{D}_{i,j}. \quad (10)$$

In SSE, the intra-class scatter matrix $\mathcal{B}_w$ and inter-class scatter matrix $\mathcal{B}_b$ are defined by:

$$\mathcal{B}_w = \sum_{c=1}^C \sum_{\mathcal{D}_{i,j} \in c} (\mathcal{D}_{i,j} - \mu_{i,j,c})(\mathcal{D}_{i,j} - \mu_{i,j,c})^T, \quad (11)$$

$$\mathcal{B}_b = \sum_{c=1}^C \omega_c (\mu_{i,j,c} - \mu_{i,j})(\mu_{i,j,c} - \mu_{i,j})^T, \quad (12)$$

where ω_c denotes the corresponding weight for class c , and $\omega_c = N_{i,j,c}$ is set in SSE. Therefore, the intra-class distance $\mathcal{H}_w$ and the inter-class distance $\mathcal{H}_b$ in subspace enhancement can be defined as below:

$$\mathcal{H}_w = \tilde{\mathbf{w}}^T \mathcal{B}_w \tilde{\mathbf{w}}, \quad (13)$$

$$\mathcal{H}_b = \tilde{\mathbf{w}}^T \mathcal{B}_b \tilde{\mathbf{w}}. \quad (14)$$

To obtain subspace enhanced features that maintain the largest inter-class distance and the smallest intra-class distance, the objective function can be defined as:

$$\arg \max_{\tilde{\mathbf{w}}} J(\tilde{\mathbf{w}}) = \frac{|\mathcal{H}_b|}{|\mathcal{H}_w|} = \frac{|\tilde{\mathbf{w}}^T \mathcal{B}_b \tilde{\mathbf{w}}|}{|\tilde{\mathbf{w}}^T \mathcal{B}_w \tilde{\mathbf{w}}|}. \quad (15)$$

For (15), we restrict the denominator to 1 and utilize the Lagrange multiplier to solve it. Then the objective function is transformed as follows:

$$\arg \min_{\tilde{\mathbf{w}}} -\tilde{\mathbf{w}}^T \mathcal{B}_b \tilde{\mathbf{w}} \quad \text{s.t.} \quad \tilde{\mathbf{w}}^T \mathcal{B}_w \tilde{\mathbf{w}} = 1, \quad (16)$$

$$J(\tilde{\mathbf{w}}) = -\tilde{\mathbf{w}}^T \mathcal{B}_b \tilde{\mathbf{w}} + \lambda(\tilde{\mathbf{w}}^T \mathcal{B}_w \tilde{\mathbf{w}} - 1). \quad (17)$$

Finally, the detailed derivation process is as follows:

$$\frac{\partial J(\tilde{\mathbf{w}})}{\partial \tilde{\mathbf{w}}} = -2 \mathcal{B}_b \tilde{\mathbf{w}} + 2\lambda \mathcal{B}_w \tilde{\mathbf{w}} = 0, \quad (18)$$

$$\mathcal{B}_b \tilde{\mathbf{w}} = \lambda \mathcal{B}_w \tilde{\mathbf{w}}, \quad (19)$$

$$\mathcal{B}_w^{-1} \mathcal{B}_b \tilde{\mathbf{w}} = \lambda \tilde{\mathbf{w}}. \quad (20)$$

Algorithm 1 Superior Subspaces Enhancement (SSE)

Require:

Input: the training set $\mathbf{X}^{train}$, the size of the search subspace $\mathcal{r}$, the multiple φ , $\varepsilon = 0.8$;

Ensure:

- 1: Generate shuffle feature $\mathbf{X}^{new}$ from $\mathbf{X}^{train}$;
- 2: Calculate the number of features in each superior subspace q by (4);
- 3: Calculate the number of random trees z by (5);
- 4: **For** i in $1, 2, \dots, z$:
- 5: Construct the random tree T^i with $\mathcal{r}$ to obtain the importance scores of features;
- 6: Select the top- q features based on the importance scores to form a new superior subspace Φ^i ;
- 7: Employ bootstrap with ε to produce subsets $\mathcal{D}_i$ in Φ^i , and then, apply LDA on $\mathcal{D}_i$ to get subspace enhanced features $\mathbf{S}^i$;
- 8: **End For**;
- 9: Construct $\mathbf{S}$ by concatenating $\mathbf{S}^i$ ($i \in \{1, 2, \dots, z\}$);

Output: subspace enhanced features $\mathbf{S}$.

The pseudo-code of SSE is presented in Algorithm 1. The superiorities of SSE include: 1) SSE effectively executes feature selection and transformation on high-dimensional features; 2) It can obtain multiple superior feature subspaces with diversity and discrimination, which alleviates effects of noise or redundant features; 3) It can extract more compact and effective features in each superior feature subspace, which enhances the expression ability of features; and 4) SSE has high flexibility since various feature transformation strategies can be combined easily.

3.2 Mixed Space Enhancement

Rotation-based ensemble method is an effective ensemble classification. Its main idea is to adopt linear combination on the original feature subset to generate new features, which encourages diversity and accuracy simultaneously [55], [83], [84]. Compared with bagging and boosting, rotation forest conducts more effective feature transformation, which has better generalization and robustness.

The main steps of the original rotation-based ensemble method are as follows. To obtain diverse training data for base classifier, the original input feature space is randomly divided into K disjoint subspaces and PCA is performed on each subspace with 75% bootstrapped samples. All the principal components of PCA are retained to preserve comprehensive information of the data. Finally, the majority voting scheme is used to obtain the final prediction.

The reason why rotation-based ensemble method can be effective is that it can make the training subsets more diverse to increase differences between base classifiers in an ensemble system. By PCA, any possibly correlated features can be orthogonally transformed into a group of linearly uncorrelated features. Meanwhile, the original features are fused to generate multivariate features. Compared to univariate-based decision model, multivariate-based decision model has more powerful representation capabilities.

However, most rotation-based feature fusion methods are performed on the original feature space, which is defective in high-dimensional data with redundant and noisy features. To construct effective feature fusion, we develop a mixed space enhancement method (MSE) based on multiscale rotation reconstruction and subspace enhanced features to generate diverse and powerful features. Compared to the original feature space, subspace enhanced features obtained from SSE are more compact and effective. Algorithm 2 describes pseudo-code of our integrated system CESE, which includes superior subspace enhancement (SSE) and mixed space enhancement (MSE). For each base classifier, the number of features in each rotation subspaces M in MSE is randomly set to one of the scales, such as 3, 5, or 7, which can effectively increase the diversity within the ensemble system.

Algorithm 2 Classifier ensemble method based on subspace enhancement (CESE)

Require:

Input: the training set $\mathbf{X}^{train}$, the size of the search subspace r , the multiple φ , $\varepsilon = 0.8$, the number of subspaces ℓ (the number of features in each rotation subspaces M , $M \in \{3, 5, 7\}$), the number of base classifiers k , the base classifier algorithm D ;

Ensure:

- 1: **For** i in 1, 2, ..., k :
- 2: Apply Algorithm 1 (SSE) based on r , φ , ε to generate the subspace enhanced features $\mathbf{S}^i$ of $\mathbf{X}^{train}$.
- 3: Randomly select one of the scales for M ;
- 4: Randomly divide the subspace enhanced features $\mathbf{S}^i$ into ℓ disjoint subspaces $\mathbf{F}_{i,j}$ ($j = \{1, 2, \dots, \ell\}$);
- 5: **For** j in 1, 2, ..., ℓ :

- 6: Construct new training set $\mathbf{X}_{i,j}$ from $\mathbf{S}^i$ based on $\mathbf{F}_{i,j}$;
 - 7: Use bootstrap to select 75% of samples from $\mathbf{X}_{i,j}$ to obtain a new training set $\mathbf{X}'_{i,j}$;
 - 8: Use PCA on $\mathbf{X}'_{i,j}$ to obtain the coefficients $\mathbf{C}_{i,j}$;
 - 9: **End For**;
 - 10: Arrange a rotation matrix $\mathbf{R}_i$ based on $\mathbf{C}_{i,j}$, which is defined as:
- $$\mathbf{R}_i = \begin{bmatrix} \mathbf{C}_{i,1}^{(1)}, \dots, \mathbf{C}_{i,1}^{(M_1)} & [0] & \dots & [0] \\ [0] & \mathbf{C}_{i,2}^{(1)}, \dots, \mathbf{C}_{i,2}^{(M_2)} & \dots & [0] \\ \vdots & \vdots & \ddots & \vdots \\ [0] & [0] & \dots & \mathbf{C}_{i,\ell}^{(1)}, \dots, \mathbf{C}_{i,\ell}^{(M_\ell)} \end{bmatrix}$$
- 11: Rearrange $\mathbf{R}_i$ to $\mathbf{R}_i^a$ concerning the original features;
 - 12: Create the new training set $\mathbf{X}_{final}^i$ by (21) or (23);
 - 13: Train the base classifier D_i by $\mathbf{X}_{final}^i$;
 - 14: **End For**;
 - 15: Use the majority voting to obtain the prediction;

Output: the classifier ensemble based on subspace enhancement $\mathbf{D}$.

TABLE 1 DIFFERENT FEATURE COMBINATIONS.

No	Method	Scheme	Equation
1	CESE_0 (only SSE)	-	SSE (X)
2	CESE (Flow based)	Flow	MSE (SSE(X))
3	CESE (Difference based)	Operation	MSE (SSE(X)) - SSE(X)
4	CESE (Sum based)	Operation	MSE (SSE(X)) + SSE(X)
5	CESE (RS based)	Operation	RS{MSE(SSE(X)), SSE(X)}
6	CESE (Concatenation)	Operation	{MSE(SSE(X)), SSE(X)}

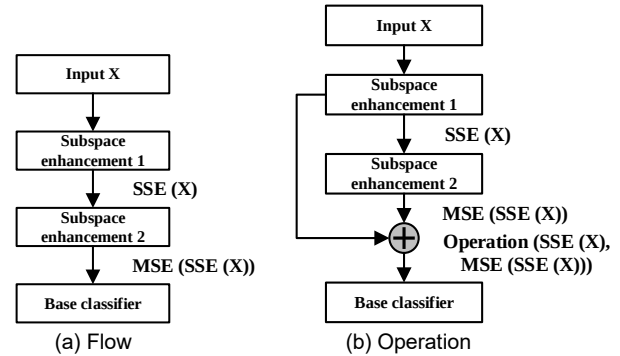


Fig. 3. The feature combination frameworks in CESE.

Since features play a key role in decision-making of classifier, we introduce multiple feature combinations for CESE to merge enhanced features generated by SSE and MSE. Although some researchers proposed feature combination methods to obtain better results [56], these methods are applied in the original input feature space directly, which are easily affected by the quality of the input feature space. Therefore, to further improve the capacity of our ensemble model, we design a variety of feature combination strategies for each base classifier. To better demonstrate the enhancement brought by different feature combinations, multiple kinds of feature combination structures are constructed as Fig. 3 and Table 1.

1) *Flow-based framework*: as shown in Fig. 3 (a), for each

base classifier in CESE, the input training data are fed into the first subspace enhancement (SSE) to obtain subspace enhanced features $SSE(\mathbf{X})$. Then subspace enhanced features are implemented feature fusion in the second subspace enhancement (MSE). Diverse mixed enhanced features $MSE(SSE(\mathbf{X}))$ are generated by multi-scale rotation reconstruction. Finally, the base classifier is trained on the mixed enhanced features $MSE(SSE(\mathbf{X}))$. The above steps are repeated k times to construct k base classifiers.

2) *Operation based framework*: for CESE with the operation framework, several operations are designed to combine subspace enhanced features $SSE(\mathbf{X})$ and mixed enhanced features $MSE(SSE(\mathbf{X}))$. As shown in Fig. 3 (b), the operation is applied on $SSE(\mathbf{X})$ and $MSE(SSE(\mathbf{X}))$. To investigate which one can achieve higher performance for CESE, *concatenation*, *sum*, *difference*, and *random subspace* operations are adopted in our CESE. Therefore, each base classifier is trained with combined features Operation ($SSE(\mathbf{X})$, $MSE(SSE(\mathbf{X}))$). Similarly, we conduct the above steps k times independently to build k base classifiers.

After subspace enhancements of CESE with flow framework, the final input of base classifier D_i is defined as:

$$\mathbf{X}_{final}^i = MSE_i(SSE_i(\mathbf{X}^{train})) , \quad (21)$$

$$\mathbf{X}_{final}^i = \mathbf{S}^i \mathbf{R}_i^a . \quad (22)$$

For our method CESE with the operation framework, the final features of base classifier D_i can be given as:

$$\mathbf{X}_{final}^i = SSE_i(\mathbf{X}^{train}) \oplus MSE_i(SSE_i(\mathbf{X}^{train})) , \quad (23)$$

$$\mathbf{X}_{final}^i = \mathbf{S}^i \oplus \mathbf{S}^i \mathbf{R}_i^a , \quad (24)$$

where $\oplus$ denotes the operation for feature combination.

Similar to most integrated systems, the final prediction result in CESE is constructed by the average voting scheme. Suppose $\mathcal{P} = \{\mathcal{P}_1, \mathcal{P}_2, \dots, \mathcal{P}_k\}$ is the confidence of prediction results from base classifiers $\mathcal{D} = \{D_1, D_2, \dots, D_k\}$. The confidence of data $\mathbf{x}_i$ is classified to be c -th class by:

$$\mathbb{C}_c(\mathbf{x}_i) = \frac{1}{k} \sum_{j=1}^k \mathcal{P}_{j,c}(\mathbf{x}_i) , \quad (25)$$

where k denotes the number of base classifiers. The final results $\mathbb{E}(\mathbf{x}_i)$ of the input data $\mathbf{x}_i$ is given by:

$$\mathbb{E}(\mathbf{x}_i) = \arg \max_{c \in \{1, 2, \dots, C\}} \mathbb{C}_c(\mathbf{x}_i) , \quad (26)$$

where C denotes the number of classes.

Advantages of MSE in CESE are as follows: 1) It can generate diverse and creative features derived from subspace enhanced features of SSE; 2) The mixed enhanced features obtained from MSE play an auxiliary role in enhancing the ability of feature expression via multi-scale feature fusion mechanism; and 3) MSE has high scalability since feature fusion based on rotation can be replaced by any operations.

4 EXPERIMENT

4.1 Dataset and Evaluation Metric

Table 2 shows 33 real-world high-dimensional data sets. These data sets are selected from various fields, which include gene expression profiles, bioinformatics data, images, and high-dimensional data from UCI. It can be seen that

most data sets are challenging because of high-dimensional features and small sample size. For example, ALL contains 72 samples with 7129 features, Glioma contains 50 samples with 4434 features, and Lymphoma contains 96 samples with 4026 features.

To evaluate performance of methods, we apply accuracy (AC) as the evaluation metric, which is defined by:

$$AC = \frac{1}{|\mathbf{X}^{test}|} \sum_{\mathbf{x}_i \in \mathbf{X}^{test}} 1\{\mathbf{y}_i^{predict} == \mathbf{y}_i^{true}\} , \quad (27)$$

where $\mathbf{X}^{test}$ is the testing set, $|\mathbf{X}^{test}|$ is the number of samples in $\mathbf{X}^{test}$, $\mathbf{y}_i^{true}$ is the true label of $\mathbf{x}_i$, and $\mathbf{y}_i^{predict}$ is the predicted label. In the experiments, we conduct five times 5-fold cross-validation to reduce the randomness and provide the accuracy in the form of mean $\pm$ standard deviation. Meanwhile, for all the comparison classifier ensemble methods, the number of base classifiers is set to 20.

TABLE 2 A SUMMARY OF DATASETS, WHERE N , M , AND C DENOTE THE NUMBERS OF SAMPLES, FEATURES, AND CLASSES.

Dataset	Rename	Source	N	M	C
Aliz_v3	D1	[87]	62	2093	4
Aliz_v30	D2	[87]	62	4026	4
ALL	D3	[90]	72	7129	2
Ar	D4	[48]	260	1260	10
Armstrong	D5	[87]	72	2194	3
Bredel	D6	[87]	50	1739	3
Carcinom	D7	[88]	174	9182	11
CIL_sub	D8	[89]	111	11340	3
Cnae_9	D9	[90]	1080	856	9
Coil	D10	[91]	360	1024	5
Colon	D11	[46]	62	2000	2
DLBCL_A	D12	[48]	141	661	3
DLBCL_B	D13	[48]	180	661	3
Dyrskjot	D14	[87]	40	1203	3
Glioma	D15	[89]	50	4434	4
Lapointe_v2	D16	[87]	110	2496	4
Leukemia	D17	[92]	248	985	6
LSVT	D18	[90]	126	310	2
Lung	D19	[48]	203	3312	4
Lymphoma	D20	[93]	96	4026	7
Mfeat	D21	[90]	2000	649	10
Novartis	D22	[94]	103	1000	4
Nutt	D23	[87]	50	1377	4
Orl	D24	[95]	210	1024	21
Pie	D25	[96]	210	2420	10
Pomeroy	D26	[97]	42	1379	5
Risinger	D27	[87]	42	1771	4
Su	D28	[87]	174	1571	10
Tomlins_v1	D29	[87]	104	2315	5
Tomlins_v2	D30	[87]	92	1288	4
Tox	D31	[98]	171	5748	4
YaleB	D32	[99]	320	1024	5
Yeoh	D33	[87]	248	2526	6

Table 3 shows a summary of three crucial parameters in CESE, which include the multiple size in SSE, the search subspace size, and feature combination frameworks. We investigate the effect of one parameter with the other parameters fixed to default values.

To make a comprehensive analysis of our CESE, we first investigate effects of important parameters for CESE and then adopt the Kappa-Error diagram to analyze the diversity and accuracy of CESE and its competitors. Finally, to

verify the effectiveness of CESE, it is compared with different mainstream classifier ensemble systems.

TABLE 3 PARAMETER SETTINGS.

No	Parameters	Default Value	Range
1	The multiple size in SSE ϕ	6x	2x, 4x, 6x, 8x, 10x
2	The search subspace size r	8	4, 6, 8, 10, 12
3	The feature combination framework	<i>concatenation</i>	<i>flow, concatenation, sum, difference, random subspace</i>

4.2 Effect of the Multiple Size in SSE

The multiple size ϕ is crucial to the success of feature transformation. To explore the effect of ϕ , we conduct experiments on ALL, DLBCL_A, DLBCL_B, and LSVT concerning different multiple sizes, as reported in Fig. 1 in the supplementary file. It is observed that a better performance can be obtained when setting the multiple size to 6x except for LSVT, on which the best performance is achieved when ϕ is 4x. A possible reason is that the number of features in LSVT is relatively small, the diversity of subspace enhanced features will be reduced when ϕ is large. However, performance of the model is still satisfactory when ϕ is set to 6x. In a small superior subspace, our SSE obtains better subspace enhanced features that are more diverse and discriminative. However, if the multiple size in SSE is extremely small, it leads to insufficient effective information, which may degrade the performance. Larger multiple size in SSE can consider more information for feature transformation. While it may be easily affected by redundant features in high-dimensional data. Meanwhile, as the multiple size in SSE becomes larger, the diversity of subspace enhanced features begins to decline. Therefore, the default value 6x of the multiple size is suitable for our proposed SSE.

4.3 Effect of the Search Subspace Size

The search subspace size r for each random tree is another key parameter. The effects of different search subspace sizes for SSE is reported in Fig. 2 in the supplementary file. It is noticed that when the multiple size is selected to 6x (default value) and the search subspace size is selected to 8, our SSE acquires better accuracy on most high-dimensional data sets except for ALL, on which the best performance is obtained when r is 4. A possible reason is that there are a lot of redundant and noisy features in ALL. A large search subspace size results in redundant features between superior subspaces, which might produce negative effects on performance of the model. From Fig. 2 in the supplementary file, when the search subspace size increases to a certain value (more than 8), performance of our method SSE drops. When the search subspace size is small (less than 8), with the increase of search subspace size, SSE can provide more effective feature selection to obtain superior feature subspaces, and performance of SSE can be gradually improved. A possible reason is that a smaller search subspace makes superior subspaces more diverse, while a larger search subspace makes superior subspaces more similar. Although a larger search subspace explores a wider range of features, it may result in the redundancy

between superior subspaces. Therefore, the default value 8 of the search subspace size is suitable for SSE.

4.4 Effect of the Feature Combination Framework

To enhance the ability of feature expression of our CESE, we design various feature combination frameworks for the subspace enhanced features SSE($\mathbf{X}$) and mixed enhanced features MSE (SSE($\mathbf{X}$)). To better investigate the effectiveness of feature combinations on CESE, several feature combination strategies are described in Fig. 3, which include the flow-based framework and the operation based framework. The comparisons of feature combinations for CESE are shown in Table 1 in the supplementary file. Compared with CESE_0 (only SSE), CESE (Flow based) obtains performance improvement on average accuracy, which proves the effectiveness of our proposed MSE. The reason is that MSE can generate diverse and creative features from subspace enhanced features of SSE, which plays an auxiliary role in CESE. Furthermore, compared to CESE_0 (only SSE), the operation based framework with *concatenation* has delivered better performance on 24 out of 33 data sets, because the operation based framework with *concatenation* preserves complete information of both SSE($\mathbf{X}$) and MSE (SSE($\mathbf{X}$)). Among operation-based frameworks, CESE with *random subspace* (RS based) achieves poor results. Although CESE (RS based) increases diversities between base classifiers by using random subspace tricks on SSE($\mathbf{X}$) and MSE (SSE($\mathbf{X}$)), it leads to the lack of sufficient feature information for each classifier. On the contrary, from Table 1 in the supplementary file, CESE (Concatenation) achieves better performance on the majority of high-dimensional data sets. Therefore, setting the operation with *concatenation* as the feature combination strategy is considered to be the most appropriate on CESE.

4.5 Diversity and Error Analysis

To analyze the diversity and accuracy results of integrated systems, we apply the Kappa-error diagram [80] on our CESE, Random Forest, and Bagging. Fig. 3 in the supplementary file shows the diversity and error of three integrated systems on Carcinom, Cll_sub, Lapointe_v2, Su, Cnae_9, Tomlins_v2, Tox, and Yeoh. Note that a high diversity of the classifier corresponds to a small Kappa value. As shown in Fig. 3 in the supplementary file, points located in the lower-left corner represent higher diversity and higher classification accuracy. An ideal integrated system should have enough diversity with low classification error. It can be seen that the diversity of Bagging is lower than that of Random Forest since the latter applies randomization process on feature and sample space to train diverse classifiers, while the former only utilizes randomization process on sample space. Besides, the error of base classifiers in Random Forest is higher than that of base classifiers in Bagging. A possible reason is that Random Forest adopts feature subspace to generate base classifiers, which may lead to information loss. Fig. 3 in the supplementary file shows that the base classifiers of CESE have high accuracy with enough diversity on different high-dimensional data sets.

4.6 Analysis for Comparative Experimental Results

To verify the effectiveness of our method, CESE is compared with mainstream ensemble learning methods which include: 1) ensemble methods based on feature space; 2) ensemble methods based on sample space; and 3) ensemble methods based on hybrid strategies.

First, for integrated systems with feature space, random forest [54], extremely randomized trees (ExtraTrees) [67], random subspace [53], and rotation forest [55] are selected as baselines. Table 4 presents classification results on all the high-dimensional data sets, from which it can be seen that our proposed CESE overcomes other competitors and obtains the best result over 29 out of 33 data sets. Specifically, CESE performs best on Tomlins_v1 (D29), Glioma (D15), and Nutt (D23) with accuracy 0.8710, 0.8040, and 0.7480, which are 8.30%, 7.60%, and 6.80% larger than the second-best results, respectively. The reason may be that these classifier ensemble methods like random forest and random subspace take each feature as an independent individual, and they do not consider a better representation ability brought by feature transformation or fusion. For methods based on random subspace, small feature subspace cannot ensure adequate effective features in each integrated system member. Although the problem of feature redundancy and noise can be alleviated by using subspace technique, it will lead to information loss which may degrade performance of classifier. To address the above limitation, superior subspaces enhancement method (SSE) is proposed in CESE, which can effectively implement feature selection and feature transformation. In SSE, multiple superior feature subspaces with diversity and discrimination are generated, which effectively alleviates effects of noise and redundant features. More importantly, to mine relationships between features and labels, the information about the labels is utilized in feature selection and feature transformation of SSE.

Second, our proposed approach CESE is compared with integrated systems based on sample space, which consist of Bagging [51], AdaBoost [57], Gradient Boosting Decision Tree (GBDT) [85], and XGBoost [86]. The comparison results are reported in Table 5. Our method CESE performs best on all the data sets. For example, the accuracy of CESE on Tox (D31), DLBCL_A (D12), and DLBCL_B (D13) are 0.8656, 0.9786, and 0.9633. It outperforms the second-best method by 13.57%, 12.92%, and 11.78%, respectively. The possible reason is that these methods adopt sample subspace to create different base classifiers, but all the feature information is retained, which is easily affected by redundant or noise features in the high-dimensional data. Meanwhile, such data are accompanied by the characteristics of small samples. Constructing a smaller sample subspace leads to an insufficient description for the sample space, which will greatly affect the accuracy of integrated system members. To construct more effective feature fusion, CESE develops a mixed space enhancement method (MSE) based

on multiscale rotation reconstruction and subspace enhanced features to generate more diverse and discriminative features. Compared with original input feature spaces, subspace enhanced features derived from SSE are more compact and effective. Besides, the feature combination in CESE further enhances the ability of feature expression.

Third, we compare CESE with classifier ensemble methods based on hybrid strategies, which include RotBoost [74], Rotated random forest (RoRF) [68], Extended space random forest (ESRF) [56], RSM-IPCS [76], and NEC [75]. The comparison results are reported in Table 6. It can be seen that our CESE outperforms its competitors over 28 out of 33 data sets. For example, CESE achieves the best classification performance on Tomlins_v1 (D29), Glioma (D15), and Tomlins_v2 (D30) with accuracy 0.8710, 0.8040, and 0.8326, which are 9.64%, 8.80%, and 8.33% larger than the second-best methods, respectively. Similarly, these ensemble methods based on hybrid strategies suffer from noise or redundant features and small-size samples. On the contrary, our CESE adopts the superior subspaces enhancement (SSE) and the mixed space enhancement method (MSE) to obtain more accurate and diverse base classifiers in an ensemble model, and it is more proper for high-dimensional data classification.

4.7 Effect of Base Classifiers

To explore the effects of base classifiers for CESE, we compare performance of CESE with different base classifiers. Fig. 4 in the supplementary file shows accuracies obtained by CESE-DT, CESE-RF, and CESE-EF. CESE-DT, CESE-RF, and CESE-EF denote CESE with decision tree, random forest and extremely randomized forest as base classifiers, respectively. Fig. 4 in the supplementary file demonstrates that CESE-EF achieves the best performance, and overcomes its competitors over 25 out of 33 data sets. Specifically, CESE-EF acquires the best results on Pomeroy (D26), Risinger (D27), and Glioma (D15) with accuracy 0.7270, 0.7340, and 0.8040, which are 5.60%, 3.40%, and 3.20% larger than the second-best classification results, respectively. A possible reason may be that the decision tree is prone to over-fitting in the construction process, which can weaken the generalization of CESE. Compared with decision tree, random forest and extremely randomized forest effectively increase the generalization by introducing the idea of ensemble learning and obtain better results. However, random forest applies randomization process on feature and sample space to build diverse base classifiers, which may lead to an insufficient description of sample space and affect the accuracy of base classifiers. Unlikely, extremely randomized forest does not use the bootstrap method on sample space to construct base classifiers. It can be observed that CESE-DT obtains satisfactory performances on Novartis (D22) and Tox (D31), while CESE-RF achieves satisfactory performance on ALL (D3) and Yeoh (D33). In summary, CESE-EF based on extremely randomized forest is preferred for high-dimensional data.

TABLE 4 COMPARISON OF DIFFERENT CLASSIFIER ENSEMBLE METHODS BASED ON FEATURE SPACE.

Datasets	RandomForest	ExtraTrees	Random Subspace	RotationForest	CESE (our method)
D1	0.7909 ± 0.0478	0.8152 ± 0.0427	0.8133 ± 0.0269	0.8623 ± 0.0303	0.9014 ± 0.0361
D2	0.8228 ± 0.0324	0.8495 ± 0.0313	0.8214 ± 0.0145	0.8604 ± 0.0293	0.9185 ± 0.0098

D3	0.9075 ± 0.0379	0.9132 ± 0.0246	0.9110 ± 0.0209	0.9414 ± 0.0115	0.9757 ± 0.0060
D4	0.9476 ± 0.0127	0.9576 ± 0.0159	0.8853 ± 0.0061	0.9823 ± 0.0030	1.0000 ± 0.0000
D5	0.9353 ± 0.0260	0.9328 ± 0.0222	0.8653 ± 0.0293	0.9407 ± 0.0109	0.9667 ± 0.0064
D6	0.8000 ± 0.0219	0.7920 ± 0.0271	0.8000 ± 0.0282	0.8360 ± 0.0265	0.8480 ± 0.0160
D7	0.8291 ± 0.0258	0.8171 ± 0.0227	0.8364 ± 0.0160	0.8587 ± 0.0149	0.9248 ± 0.0060
D8	0.6717 ± 0.0331	0.6696 ± 0.0270	0.7063 ± 0.0310	0.7566 ± 0.0219	0.7654 ± 0.0125
D9	0.9090 ± 0.0058	0.9229 ± 0.0068	0.9311 ± 0.0036	0.8712 ± 0.0056	0.9277 ± 0.0019
D10	0.9961 ± 0.0022	0.9966 ± 0.0027	0.9822 ± 0.0132	0.9938 ± 0.0011	0.9950 ± 0.0020
D11	0.7980 ± 0.0209	0.8095 ± 0.0185	0.7895 ± 0.0235	0.7909 ± 0.0242	0.8300 ± 0.0217
D12	0.9023 ± 0.0236	0.9148 ± 0.0308	0.8438 ± 0.0261	0.9109 ± 0.0187	0.9786 ± 0.0045
D13	0.8600 ± 0.0133	0.8911 ± 0.0151	0.8244 ± 0.0174	0.9288 ± 0.0200	0.9633 ± 0.0083
D14	0.7600 ± 0.0604	0.7900 ± 0.0406	0.7650 ± 0.0254	0.8050 ± 0.0400	0.8650 ± 0.0254
D15	0.6920 ± 0.0515	0.7080 ± 0.0515	0.6880 ± 0.0370	0.7280 ± 0.0627	0.8040 ± 0.0195
D16	0.7654 ± 0.0194	0.7872 ± 0.0302	0.7763 ± 0.0451	0.8072 ± 0.0156	0.8563 ± 0.0089
D17	0.9580 ± 0.0081	0.9588 ± 0.0092	0.9377 ± 0.0093	0.9732 ± 0.0075	0.9886 ± 0.0029
D18	0.8188 ± 0.0151	0.8107 ± 0.0116	0.7920 ± 0.0220	0.8222 ± 0.0356	0.8759 ± 0.0142
D19	0.9089 ± 0.0040	0.9167 ± 0.0057	0.9216 ± 0.0114	0.9344 ± 0.0105	0.9535 ± 0.0065
D20	0.8153 ± 0.0348	0.8483 ± 0.0224	0.8291 ± 0.0331	0.8547 ± 0.0129	0.8562 ± 0.0241
D21	0.9805 ± 0.0015	0.9818 ± 0.0011	0.9778 ± 0.0019	0.9798 ± 0.0007	0.9898 ± 0.0002
D22	0.9613 ± 0.0126	0.9627 ± 0.0072	0.9368 ± 0.0278	0.9713 ± 0.0051	0.9842 ± 0.0047
D23	0.6400 ± 0.0726	0.6800 ± 0.0456	0.5880 ± 0.0271	0.6600 ± 0.0219	0.7480 ± 0.0430
D24	0.9676 ± 0.0110	0.9847 ± 0.0035	0.8914 ± 0.0145	0.9552 ± 0.0122	0.9780 ± 0.0093
D25	0.9638 ± 0.0088	0.9857 ± 0.0085	0.8980 ± 0.0267	0.9942 ± 0.0035	1.0000 ± 0.0000
D26	0.6450 ± 0.0620	0.6810 ± 0.0558	0.5960 ± 0.0695	0.6970 ± 0.0838	0.7270 ± 0.0278
D27	0.6780 ± 0.0560	0.6920 ± 0.0482	0.5900 ± 0.0330	0.6820 ± 0.0280	0.7340 ± 0.0406
D28	0.8538 ± 0.0090	0.8495 ± 0.0093	0.8630 ± 0.0124	0.8448 ± 0.0201	0.9024 ± 0.0093
D29	0.7303 ± 0.0267	0.7550 ± 0.0236	0.7330 ± 0.0394	0.7880 ± 0.0163	0.8710 ± 0.0212
D30	0.7011 ± 0.0550	0.6897 ± 0.0420	0.6991 ± 0.0366	0.7742 ± 0.0233	0.8326 ± 0.0139
D31	0.7053 ± 0.0155	0.7158 ± 0.0192	0.7238 ± 0.0336	0.8245 ± 0.0098	0.8656 ± 0.0082
D32	0.9937 ± 0.0027	0.9862 ± 0.0085	0.9731 ± 0.0067	0.9962 ± 0.0012	0.9950 ± 0.0031
D33	0.6582 ± 0.0332	0.6700 ± 0.0220	0.7983 ± 0.0093	0.7974 ± 0.0095	0.8356 ± 0.0155
Number of wins	0/33	2/33	1/33	1/33	29/33

TABLE 5 COMPARISON OF DIFFERENT CLASSIFIER ENSEMBLE METHODS BASED ON SAMPLE SPACE.

Datasets	Bagging	AdaBoost	GBDT	XGBoost	CESE (our method)
D1	0.8376 ± 0.0277	0.4633 ± 0.0732	0.7214 ± 0.0276	0.7495 ± 0.0654	0.9014 ± 0.0361
D2	0.8242 ± 0.0466	0.5142 ± 0.0669	0.7566 ± 0.0330	0.7319 ± 0.0235	0.9185 ± 0.0098
D3	0.9225 ± 0.0274	0.9364 ± 0.0214	0.8607 ± 0.0116	0.9146 ± 0.0258	0.9757 ± 0.0060
D4	0.9238 ± 0.0134	0.2353 ± 0.0170	0.8315 ± 0.0220	0.9053 ± 0.0062	1.0000 ± 0.0000
D5	0.9153 ± 0.0175	0.8496 ± 0.0432	0.8639 ± 0.0330	0.9053 ± 0.0245	0.9667 ± 0.0064
D6	0.8160 ± 0.0233	0.7400 ± 0.0178	0.7760 ± 0.0624	0.7800 ± 0.0419	0.8480 ± 0.0160
D7	0.8331 ± 0.0214	0.4508 ± 0.0269	0.8126 ± 0.0258	0.8117 ± 0.0232	0.9248 ± 0.0060
D8	0.7242 ± 0.0392	0.6406 ± 0.0298	0.7080 ± 0.0190	0.7242 ± 0.0404	0.7654 ± 0.0125
D9	0.8827 ± 0.0065	0.3750 ± 0.0423	0.9038 ± 0.0025	0.8677 ± 0.0035	0.9277 ± 0.0019
D10	0.9838 ± 0.0091	0.5305 ± 0.0112	0.9611 ± 0.0074	0.9650 ± 0.0048	0.9950 ± 0.0020
D11	0.7909 ± 0.0321	0.7285 ± 0.0246	0.7204 ± 0.0404	0.8047 ± 0.0265	0.8300 ± 0.0217
D12	0.8494 ± 0.0119	0.7163 ± 0.0379	0.8329 ± 0.0369	0.8494 ± 0.0162	0.9786 ± 0.0045
D13	0.8455 ± 0.0330	0.7644 ± 0.0143	0.8055 ± 0.0272	0.8366 ± 0.0184	0.9633 ± 0.0083
D14	0.7900 ± 0.0339	0.7000 ± 0.0273	0.6550 ± 0.0731	0.6550 ± 0.0187	0.8650 ± 0.0254
D15	0.6720 ± 0.0890	0.5400 ± 0.0282	0.6120 ± 0.0370	0.6920 ± 0.0370	0.8040 ± 0.0195
D16	0.7800 ± 0.0218	0.5636 ± 0.0304	0.7581 ± 0.0400	0.7963 ± 0.0347	0.8563 ± 0.0089
D17	0.9539 ± 0.0067	0.7828 ± 0.0178	0.9115 ± 0.0168	0.9288 ± 0.0147	0.9886 ± 0.0029
D18	0.8233 ± 0.0152	0.7936 ± 0.0214	0.7919 ± 0.0252	0.8270 ± 0.0177	0.8759 ± 0.0142
D19	0.9235 ± 0.0112	0.8373 ± 0.0324	0.9329 ± 0.0065	0.9056 ± 0.0164	0.9535 ± 0.0065
D20	0.8130 ± 0.0318	0.5191 ± 0.0497	0.7502 ± 0.0253	0.8003 ± 0.0219	0.8562 ± 0.0241
D21	0.9717 ± 0.0018	0.2480 ± 0.0197	0.9615 ± 0.0034	0.9717 ± 0.0023	0.9898 ± 0.0002
D22	0.9675 ± 0.0085	0.6743 ± 0.0266	0.9418 ± 0.0081	0.9400 ± 0.0169	0.9842 ± 0.0047
D23	0.6680 ± 0.0545	0.3760 ± 0.0496	0.4680 ± 0.0370	0.5640 ± 0.0233	0.7480 ± 0.0430
D24	0.9285 ± 0.0225	0.1733 ± 0.0166	0.7552 ± 0.0115	0.8466 ± 0.0101	0.9780 ± 0.0093
D25	0.9514 ± 0.0114	0.2228 ± 0.0035	0.8495 ± 0.0093	0.8942 ± 0.0110	1.0000 ± 0.0000
D26	0.6270 ± 0.0400	0.3810 ± 0.0862	0.6040 ± 0.0511	0.6040 ± 0.0575	0.7270 ± 0.0278
D27	0.6470 ± 0.0222	0.4990 ± 0.0288	0.6230 ± 0.0599	0.5980 ± 0.0331	0.7340 ± 0.0406
D28	0.8513 ± 0.0131	0.4176 ± 0.0334	0.8077 ± 0.0117	0.8577 ± 0.0175	0.9024 ± 0.0093
D29	0.7763 ± 0.0151	0.4696 ± 0.0352	0.6736 ± 0.0365	0.7360 ± 0.0323	0.8710 ± 0.0212
D30	0.7075 ± 0.0238	0.4560 ± 0.0798	0.6111 ± 0.0357	0.7195 ± 0.0610	0.8326 ± 0.0139
D31	0.7299 ± 0.0134	0.4048 ± 0.0287	0.6525 ± 0.0264	0.7297 ± 0.0322	0.8656 ± 0.0082
D32	0.9856 ± 0.0057	0.2925 ± 0.0129	0.9662 ± 0.0128	0.9893 ± 0.0064	0.9950 ± 0.0031
D33	0.7910 ± 0.0159	0.4639 ± 0.0367	0.7804 ± 0.0100	0.8124 ± 0.0179	0.8356 ± 0.0155
Number of wins	0/33	0/33	0/33	0/33	33/33

TABLE 6 COMPARISON OF DIFFERENT CLASSIFIER ENSEMBLE METHODS BASED ON HYBRID STRATEGIES.

Datasets	RotBoost	RoRF	ESRF	RSM-IPCS	NEC	CESE (our method)
D1	0.7619 ± 0.0513	0.8709 ± 0.0299	0.7842 ± 0.0304	0.8033 ± 0.0388	0.8461 ± 0.0095	0.9014 ± 0.0361
D2	0.7990 ± 0.0518	0.8928 ± 0.0459	0.8104 ± 0.0267	0.8085 ± 0.0279	0.8314 ± 0.0332	0.9185 ± 0.0098
D3	0.9578 ± 0.0156	0.9196 ± 0.0216	0.9092 ± 0.0226	0.8832 ± 0.0283	0.9489 ± 0.0103	0.9757 ± 0.0060
D4	0.4276 ± 0.0368	0.9876 ± 0.0066	0.9669 ± 0.0096	0.9492 ± 0.0122	0.9515 ± 0.0143	1.0000 ± 0.0000
D5	0.9442 ± 0.0153	0.9617 ± 0.0050	0.9410 ± 0.0202	0.9267 ± 0.0138	0.9353 ± 0.0118	0.9667 ± 0.0064
D6	0.8360 ± 0.0344	0.7960 ± 0.0366	0.8040 ± 0.0407	0.8040 ± 0.0320	0.8240 ± 0.0080	0.8480 ± 0.0160
D7	0.5254 ± 0.0187	0.8588 ± 0.0225	0.8017 ± 0.0193	0.8622 ± 0.0211	0.8342 ± 0.0200	0.9248 ± 0.0060
D8	0.7475 ± 0.0333	0.7203 ± 0.0056	0.6378 ± 0.0188	0.6792 ± 0.0211	0.7732 ± 0.0154	0.7654 ± 0.0125
D9	0.4407 ± 0.0282	0.9092 ± 0.0029	0.9009 ± 0.0067	0.9127 ± 0.0032	0.8931 ± 0.0047	0.9277 ± 0.0019
D10	0.9277 ± 0.0094	0.9972 ± 0.0017	0.9938 ± 0.0027	0.9894 ± 0.0117	0.9933 ± 0.0022	0.9950 ± 0.0020
D11	0.8038 ± 0.0156	0.7790 ± 0.0186	0.7595 ± 0.0449	0.7961 ± 0.0112	0.7952 ± 0.0067	0.8300 ± 0.0217
D12	0.8993 ± 0.0070	0.9617 ± 0.0073	0.9204 ± 0.0104	0.9023 ± 0.0206	0.9191 ± 0.0104	0.9786 ± 0.0045
D13	0.9166 ± 0.0105	0.9422 ± 0.0044	0.8955 ± 0.0196	0.8677 ± 0.0357	0.9000 ± 0.0092	0.9633 ± 0.0083

D14	0.7900 ± 0.0561	0.7850 ± 0.1113	0.7100 ± 0.0374	0.8400 ± 0.0644	0.8050 ± 0.0620	0.8650 ± 0.0254
D15	0.5400 ± 0.0632	0.7160 ± 0.0387	0.6640 ± 0.0463	0.6960 ± 0.0407	0.6800 ± 0.0489	0.8040 ± 0.0195
D16	0.7363 ± 0.0190	0.8109 ± 0.0145	0.7618 ± 0.0089	0.7945 ± 0.0234	0.7981 ± 0.0378	0.8563 ± 0.0089
D17	0.8565 ± 0.0171	0.9749 ± 0.0052	0.9653 ± 0.0041	0.9684 ± 0.0109	0.9636 ± 0.0162	0.9886 ± 0.0029
D18	0.8427 ± 0.0060	0.8380 ± 0.0081	0.8268 ± 0.0091	0.7792 ± 0.0329	0.8236 ± 0.0217	0.8759 ± 0.0142
D19	0.8963 ± 0.0147	0.9143 ± 0.0039	0.9098 ± 0.0108	0.9150 ± 0.0099	0.9330 ± 0.0038	0.9535 ± 0.0065
D20	0.7523 ± 0.0346	0.7982 ± 0.0144	0.7916 ± 0.0292	0.8213 ± 0.0156	0.8042 ± 0.0257	0.8562 ± 0.0241
D21	0.5947 ± 0.0206	0.9856 ± 0.0017	0.9798 ± 0.0013	0.9818 ± 0.0014	0.9774 ± 0.0003	0.9898 ± 0.0002
D22	0.8708 ± 0.0421	0.9805 ± 0.0063	0.9670 ± 0.0133	0.9630 ± 0.0145	0.9687 ± 0.0111	0.9842 ± 0.0047
D23	0.6440 ± 0.0344	0.7440 ± 0.0320	0.5600 ± 0.0252	0.6240 ± 0.0463	0.6320 ± 0.0515	0.7480 ± 0.0430
D24	0.5028 ± 0.0337	0.9847 ± 0.0063	0.9638 ± 0.0102	0.9504 ± 0.0118	0.9219 ± 0.0115	0.9780 ± 0.0093
D25	0.6628 ± 0.0416	0.9933 ± 0.0048	0.9866 ± 0.0069	0.9704 ± 0.0055	0.9676 ± 0.0069	1.0000 ± 0.0000
D26	0.5820 ± 0.0541	0.7550 ± 0.0144	0.6610 ± 0.0715	0.6530 ± 0.0525	0.7110 ± 0.0401	0.7270 ± 0.0278
D27	0.6830 ± 0.0374	0.6860 ± 0.0259	0.7060 ± 0.0539	0.6420 ± 0.0390	0.6620 ± 0.0273	0.7340 ± 0.0406
D28	0.4907 ± 0.0239	0.8749 ± 0.0198	0.8546 ± 0.0119	0.8730 ± 0.0144	0.8502 ± 0.0151	0.9024 ± 0.0093
D29	0.6883 ± 0.0326	0.7746 ± 0.0159	0.7230 ± 0.0296	0.7573 ± 0.0135	0.7430 ± 0.0380	0.8710 ± 0.0212
D30	0.6315 ± 0.0309	0.7442 ± 0.0155	0.7000 ± 0.0187	0.7151 ± 0.0330	0.7493 ± 0.0354	0.8326 ± 0.0139
D31	0.6069 ± 0.0110	0.7836 ± 0.0293	0.6784 ± 0.0338	0.7755 ± 0.0139	0.7284 ± 0.0213	0.8656 ± 0.0082
D32	0.8756 ± 0.0219	0.9981 ± 0.0015	0.9956 ± 0.0025	0.9875 ± 0.0019	0.9900 ± 0.0045	0.9950 ± 0.0031
D33	0.5995 ± 0.0194	0.7602 ± 0.0075	0.7380 ± 0.0117	0.7676 ± 0.0156	0.8009 ± 0.0155	0.8356 ± 0.0155
Number of wins	0/33	4/33	0/33	0/33	1/33	28/33

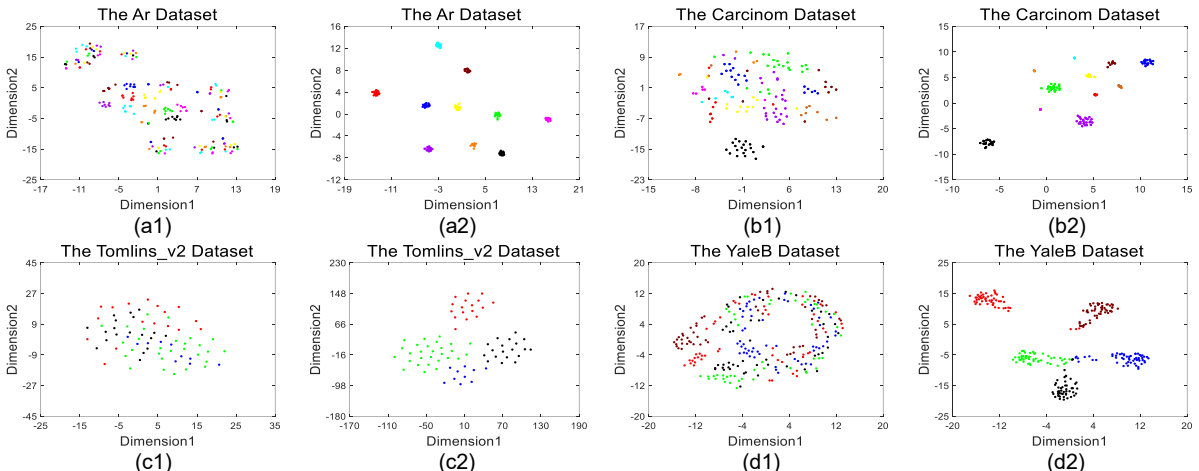


Fig. 4. Visualization of feature distributions by using t-SNE. Different colors represent different classes. All the sub-figures marked "1" or "2" denote the original features or the features obtained by our method.

TABLE 7
COMPARISON AMONG DIFFERENT CLASSIFIER ENSEMBLE
METHODS IN TERMS OF THE COMPUTATIONAL COMPLEXITY

Algorithm	Computational complexity
CESE	$O(lmu + mu^3 + lm \log(l))$
RotationForest	$O(lmu + mu^3 + lm \log(l))$
GBDT	$O(lm \log(l))$
RandomForest	$O(lm \log(l))$

TABLE 8 A SUMMARY OF LARGE DATASETS.

Dataset	Source	N	M	C
Madelon	[90]	2600	500	2
ISOLET	[90]	7797	617	26
Seizure	[90]	11500	179	5

4.8 Visualization of Our Proposed Method

To show the effectiveness of our method for high-dimensional data intuitively, we first obtain the enhanced features learnt by CESE and perform t-SNE [101] to visualize the distributions of features in Fig. 4. From Ar (Fig. 4 (a1)), Carcinom (Fig. 4 (b1)), Tomlins_v2 (Fig. 4 (c1)), and YaleB (Fig. 4 (d1)), it is observed that the feature distributions of high-dimensional data in the original feature space are irregular and chaotic. It is hardly possible to classify these data in the original feature space directly. However, after using SSE and MSE, samples between different classes can

be easily distinguished, as shown in Fig. 4 (a2), Fig. 4 (b2), Fig. 4 (c2), and Fig. 4 (d2). The reason is that CESE performs feature selection and transformation for high-dimensional data effectively, which can generate effective enhanced features. Visualization results indicate the effectiveness of our method for high-dimensional data classification.

4.9 Complexity Analysis

The time complexity T_{CESE} of CESE can be defined by:

$$T_{\text{CESE}} = T_{\text{SSE}} + T_{\text{MSE}} + T_{\text{BC}}, \quad (28)$$

where T_{SSE} , T_{MSE} , and T_{BC} denote the costs for the superior subspaces enhancement, the mixed space enhancement, and the base classifiers, respectively. T_{SSE} is related to the size of training data l , the number of features m , the size of each superior subspace q , the size of the search subspace τ , and the number of sub-classifiers k , as defined by:

$$T_{\text{SSE}} = O(k \cdot \frac{m}{q} \cdot (l \cdot q \cdot t + t^3 + l \cdot \log(l) \cdot \tau)), \quad (29)$$

where $t = \min(l, q)$. T_{MSE} is related to the size of training data l , the number of features m , the multiple ϕ , the number of features in each rotation subspaces M , and the number of sub-classifiers k as follows:

$$T_{\text{MSE}} = O(k \cdot \frac{m}{\phi \cdot M} \cdot (l \cdot M \cdot u + u^3)), \quad (30)$$

where $u = \min(l, M)$. In CESE, the tree model is applied as

the base classifier. Thus, T_{BC} is related to the size of training data l , the number of features m , the multiple ϕ , and the number of sub-classifiers k as follows:

$$T_{BC} = O(k \cdot l \cdot \frac{m}{\phi} \cdot \log(l)). \quad (31)$$

Since k , q , ϕ , r and M are smaller than l and m , and the sizes of q and M are at the same level commonly, the complexity of our CESE T_{CESE} can be denoted by $O(lmu + mu^3 + lm \log(l))$ approximately. The complexity of different classifier ensemble methods is presented in Table 7.

TABLE 9 COMPARISON OF DIFFERENT CLASSIFIER ENSEMBLE METHODS ON LARGE DATASETS.

Datasets	Random Forest	ExtraTrees	Random Subspace	Rotation Forest	Bagging	AdaBoost	GBDT	CESE (our method)
Madelon	0.6526	0.6053	0.8262	0.8270	0.8038	0.6100	0.7354	0.8657
ISOLET	0.9227	0.9270	0.9292	0.9356	0.9115	0.1530	0.8916	0.9584
Seizure	0.6222	0.6277	0.6364	0.6773	0.6286	0.4220	0.4965	0.7354

TABLE 9 (CONTINUED).

Datasets	XGBoost	RotBoost	RoRF	ESRF	RSM-IPCS	NEC	CESE (our method)
Madelon	0.7354	0.6261	0.6816	0.6260	0.7906	0.8056	0.8657
ISOLET	0.9075	0.3421	0.9490	0.9258	0.9301	0.9251	0.9584
Seizure	0.4507	0.4932	0.7329	0.6612	0.7010	0.7001	0.7354

5 CONCLUSION

In this paper, a new classifier ensemble method based on subspace enhancement is proposed for high-dimensional data classification. To generate accurate and diverse base classifiers, we first design a superior subspace enhancement scheme (SSE) for performing feature selection and transformation. In SSE, multiple superior feature subspaces with diversity and discrimination are extracted, which can enhance the expression ability of features. Meanwhile, it has high generality and expansibility. Second, to construct more effective feature fusion, we develop a mixed space enhancement method (MSE) based on multiscale rotation reconstruction and subspace enhanced features of SSE to obtain diverse and powerful features. Thirdly, to further improve the capability of our ensemble model, we design several feature combination strategies for the enhanced features of SSE and MSE. Experimental results on high-dimensional data sets indicate that our CESE outperforms mainstream integrated systems.

In this paper, we use the data sets with continuous features only, and focus on dealing with continuous features in the design process of the model. It means that the limitation of our method lies in that it cannot be directly applied to complex data which contain categorical features and missing values. Therefore, in future work, we shall consider improving CESE to handle features of any type. Additionally, we will develop an adaptive scheme to perform feature fusion and combination to optimize the classifier ensemble framework.

ACKNOWLEDGMENT

This work was supported in part by the National Key Research and Development Program of China under Grant 2019YFA0706200 and 2019YFB1703600, in part by the Key Research and Development Program of Guang Dong Province under Grant 2018B010107002.

4.10 Experiments on Large Datasets

We further conduct comparative experiments on large datasets shown in Table 8, which have more than 2000 samples. The comparative experimental results are shown in Table 9, from which it is observed that our CESE achieves the best performance and outperforms its competitors. For example, the accuracy of CESE on Madelon, ISOLET, and Seizure is 0.8657, 0.9584, and 0.7354, which are 3.87%, 0.94%, and 0.25% larger than the second-best methods, respectively. This indicates that our CESE is suitable for larger-scale data sets.

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